

SMTP/EMAIL Functions on The TMS Mainframe

SMTP and Sendmail are standard applications provided with the z/OS Communications Server. The selection of the product configuration is largely dependent on the designers criteria at SYSGEN Time. SMTP and SENDMAIL are two of the TCP/IP application protocols dealing with electronic mail. Two mail functions are supported on z/OS which are Simple Mail Transport Protocol (SMTP) in the MVS environment, and Sendmail as a z/OS UNIX component. The various standards that define the protocols for sending electronic mail are numerous part of the Request For Comment (RFC) numbers RFC 11,821-822 that represent these standards and their terminology. While the topic of email can be very complex, it is advisable that the reader familiarize themselves with these documents in order minimize any general misconceptions and confusion. As a standards, one RFC 11 is base on the formatting of the mail messages, the other two RFC's RFC 821/822 describes the configuration syntax and definitions of each service, their client/server protocol.

SMTP

The z/OS SMTP server can be set up to distribute mail on an IP network. It can also perform the role of a gateway server transferring mail between the IP network and a local NJE network. It can also be configured to send unresolved non-local mail or all non-local mail to a Mail Transfer Agent (MTA) or relay server.

SMTP on the TMS mainframe is configured as an open mail relay client. From a SMTP standpoint, the target server has no predefined knowledge of the Relay clients capabilities. If configured as such, this will allow anyone on the TMS Intranet to send e-mail through it. Any email received by the z/OS SMTP server that belongs to a valid user will forward to the predefined destination without the aid any authentication. This open relay configuration once upon a time was the default configuration in many Internet mail servers. However due to recent security concerns open mail relays have become unpopular due to their exploitation by spammers and worms. To provide addition protection many former open relays were closed, or were placed on blacklists by other servers. Since TMS's mainframe is located behind a firewall this is not an immediate security concern.

The open relay configuration setup enables NJE users to send and receive mail to and from users on TCP networks. In various implementations, there is a possibility to exchange mail between the TCP/IP SMTP mailing system and the local JES NJE spools. This type of SMTP server function is called a mail gateway or mail bridge. Sending mail through a mail gateway can alter the end-to-end delivery specification, because SMTP will only guarantee delivery to the mail-gateway host, not to the real destination host, which is located beyond the TCP/IP network. When a mail gateway is used, the SMTP end-to-end transmission is host-to-gateway, gateway-to-host, or gateway-to-gateway. The behavior beyond the gateway is not defined by SMTP.

When a client originates outbound mail destined for an unknown domain (unknown to the system where the SMTP server is running), the SMTP server can optionally forward the mail to a relay server that will be able to use special SMTP DNS lookup records to determine how and where to forward the otherwise undeliverable mail. In addition, the SMTP server can be configured to forward all *out of domain* outbound mail to a relay server

Dependencies

z/OS SMTP provides a method of originating and receiving electronic mail through the TCP/IP network using the mainframe. The use of a relay server requires a connection over the IP network, having permission to access the relay server and to send forwarded mail to it. There must be special DNS records, called Mail Exchange (MX) records, on the DNS servers in order for the SMTP server to determine the next hop, or the next relay server in the forwarding path. Once mail is delivered to a relay server, that mail is consider

delivered by the SMTP server that forwarded the mail. It then becomes the responsibility of the relay server. If the relay server cannot be contacted, then the SMTP server must have a method and a procedure for storing and managing undeliverable mail. The sender must be notified, and attempts to retry delivery must be made until successful delivery is achieved, or until retry attempts are exhausted.

Sendmail and popper:

Sendmail is an industry standard mail application that is widely used on the Internet. The sendmail application included with the z/OS V1R7.0 Communications Server is based on sendmail version 8.12.1. Popper is a Post Office Protocol 3 (POP3) mail delivery server. It allows a remote user to use a remote mail user agent (MUA) to read, compose, and manage e-mail that has been delivered to z/OS.

MTA, MUA, and MDA

When discussing sendmail and popper, other books refer to Mail Transfer Agents (MTA), Mail User Agents (MUA), and Mail Delivery Agents (MDA). Let us take a moment to define these acronyms:

- Mail Transfer Agent (MTA)
This is any application that sends a prepared e-mail message to a remote MTA. Sendmail, Microsoft Exchange, and Lotus Domino® are examples of MTAs.
- Mail User Agent (MUA)
This is any application that allows a user to read, compose, and manage e-mail. Lotus Notes, Microsoft Outlook®, and Netscape Navigator are all applications that have MUA capabilities.
- Mail Delivery Agent (MDA)
This is any application that sits between a MUA and MTU and manages delivering a message from a MTA to an MUA. A MDA is an optional piece of an e-mail system. Popper is an example of an MDA.

Dependencies

The sendmail configuration file

The sendmail configuration file is read and parsed by the sendmail program every time sendmail starts. It lists the locations of important files and specifies the default parameter values to be used within the endmail program. The parameters within the configuration file define sendmail's behavior and contain rules and rule sets for tasks such as rewriting the mailing address.

M4 preprocessor

The m4 macro processor is a front-end processor for many programming languages. Besides replacing one string of text with another, the m4 macro processor provides the following features:

Alias file

The alias file is used to convert an alias name to another recipient's name. A large number of names could potentially be listed in an alias file. In order to efficiently deal with a potentially large alias file, sendmail uses an alias database file created from a alias file. The database version of the alias file significantly improves lookup speed.

The statistics file

The sendmail.st statistics file is used by sendmail to record the number and sizes of all incoming and outgoing mail messages handled by each delivery agent. Statistics are kept for each of the following delivery agents:

The queue directory

Sendmail will create a queue directory for you the first time sendmail is run. The location of

the queue directory is specified in the sendmail configuration file. If a mail message cannot be transmitted because of certain reasons, sendmail stores it in the queue directory until the message can be transmitted successfully.